RHoMIS
Rural Household Multi-Indicator Survey

Data Visualization & Analytics
DVA Capstone 2 | 28 April 2026

# Food Security Drivers in Smallholder Farming Households

*A Multi-Country Analysis Using the RHoMIS Dataset*

54,873 households · 35 countries · 2015–2023

**Project Details**

| | |
|---|---|
| Sector | Agriculture / Rural Development |
| Team ID | Section C, Group 7 |
| Institute | Newton School of Technology |
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| Submitted | 28 April 2026 |


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| | |
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**GitHub Repository**

https://github.com/mahir-m01/Section-C_G-7_RHoMIS-Analytics

**Tableau Public Dashboard**

https://public.tableau.com/app/profile/rajdeep.sanyal/viz/RHoMISDASHBOARD

## 1. Executive Summary

| FOOD SHORTAGE RATE | IRRIGATION ACCESS GAP |
|---|---|
| **67.7%** | **16.3pp** |
| 32,097 of 47,399 households<br>Median duration: 3 months | Rainfed 71.5% vs. irrigated 55.2%<br>Only 20.2% of farms irrigated |

| LAND PRODUCTIVITY GAP | INCOME QUINTILE GAP |
|---|---|
| **29.1pp** | **19.8pp** |
| Lowest quartile: 83% shortage<br>Highest quartile: 54% shortage | Bottom quintile vs. top quintile<br>food shortage rate difference |

**Problem.** Smallholder farming households across 35 countries face chronic food insecurity. Development organisations lack clear evidence on which farm profiles and regions carry the highest risk, making it difficult to direct limited budgets toward the households that need them most.

**Approach.** The Rural Household Multi-Indicator Survey (RHoMIS) — 54,873 households, 2015–2023 — was cleaned using a Python ETL pipeline, then analysed through exploratory analysis, hypothesis testing, regression, and clustering. Findings are presented through a four-dashboard Tableau system.

**Key Findings.**

- Two in three surveyed households reported food shortage — confirming this is the structural norm, not an exception.
- Irrigation access creates a 16.3pp gap in food shortage rates; only 20.2% of households have it.
- Land productivity quartile gap (29.1pp) and income quintile gap (19.8pp) identify the structural roots of vulnerability.
- Low education and large household size compound every other risk factor.

**Key Recommendations.**

- Expand irrigation in the low-income rainfed cluster (14,330 households, 78.7% shortage, 0.3% irrigation).
- Direct productivity support to Q1 and Q2 land productivity households — the largest structural gap.
- Combine income diversification with basic education access for the bottom income quintile.

## 2. Sector & Business Context

Smallholder agriculture supports roughly 500 million farm families globally and produces the majority of food consumed in developing economies. Despite this role, smallholder households remain the most food-insecure population segment — caught in a cycle where low farm productivity limits income, which in turn limits investment, perpetuating vulnerability.

The decision-makers this analysis serves are national agricultural policy teams who allocate extension service budgets, and international development programme officers at organisations such as CGIAR and IFAD who design multi-country interventions. Both face the same question: *where do we act first, and what do we fix?*

The business value is precision targeting. Without evidence on which profiles carry the highest risk, interventions are spread thin. A country-level vulnerability map backed by household-level analysis concentrates resources on high-risk populations — reducing waste and improving measurable impact.

## 3. Problem Statement & Objectives

> Which regions and farm profiles across 35 countries are most vulnerable to food insecurity and low farm income — and which combination of factors (farm size, land productivity, crop diversity, irrigation access, household demographics, and income position) drives those outcomes?

**In scope:** food insecurity via self-reported shortage flag; structural driver analysis; household vulnerability segmentation; country-level risk mapping; gender and resource control dimensions.

**Out of scope:** causal inference; year-over-year trend analysis (cross-sectional data); sub-regional disaggregation (region names unstandardised); HFIAS frequency score (87.5% missing).

**Success criteria:** all key findings supported by formal statistical tests; Tableau dashboard navigable by a non-technical user without assistance; recommendations map to actionable structural factors.

## 4. Data Description

**Source.** Rural Household Multi-Indicator Survey (RHoMIS). Published on Harvard Dataverse by Gorman, Hammond, Frelat et al. — doi.org/10.7910/DVN/WS38SA, Version 1, January 30, 2025. CC0 1.0 Public Domain.

| File | Rows | Cols | Description |
|---|---|---|---|
| `raw.csv.gz` | 54,873 | 1,599 | Original survey data — never edited |
| `rhomis_cleaned.csv` | 54,873 | 77 | ETL pipeline output |
| `rhomis_tableau_ready.csv` | 54,873 | 46 | Final analysis-ready export |

| Column | Type | Miss. | Description |
|---|---|---|---|
| `food_shortage_flag` | Binary | 13.6% | Primary outcome. Global rate: 67.7% |
| `irrigated_flag` | Binary | 8.6% | Irrigation access. Coverage: 20.2% |
| `land_productivity_kg_pe` | Float | 12.9% | Harvest/ha. Extreme right skew — median only |
| `country_income_rank_pct` | Float | 7.1% | Within-country income percentile 0–1 |
| `crop_diversity_count` | Integer | 0.0% | Distinct crop types, 0–10. Median: 2 |
| `vulnerability_profile_k4` | String | 23.8% | k-means cluster label, 4 profiles |
| `fies_yes_count` | Float | 54.3% | FIES score 0–8. 22 of 35 countries only |

**Limitations.** Income, land productivity, and harvest fields carry extreme right skew (skewness >100) — raw values are never plotted on continuous axes; median or log transforms are used throughout. Year distribution is uneven (2018 = 22% of sample; 2015 = 2%). HFIAS shows 100% food-secure in many countries — instrument deployment artefact, not genuine outcome; excluded from primary analysis. Philippines and Gambia: zero valid food shortage responses — excluded from shortage rate charts.

## 5. Data Cleaning & ETL Pipeline

*Reference: notebooks/02_cleaning.ipynb committed to GitHub*

All cleaning and transformation steps were executed in Python via `scripts/etl_pipeline.py`. Notebooks 01 and 02 document and verify the same logic.

The pipeline narrows 1,599 raw columns to 60 working columns across identity, demographics, farm structure, crops, income, and food security. Country names are standardised to lowercase, correcting a known misspelling of "philippines." Age fields in `age_malehead` and `age_femalehead` recorded birth years in some survey waves — the pipeline detects these using each row's own survey year and converts them to ages, treating values implying age above 110 as missing.

Household size was rebuilt from scratch. The original `count_people` field is 49.4% missing, so `household_size_derived` was computed by summing eight age-band columns, yielding zero missing values. Land area was converted from local units (bigha, cuerda, etc.) using a 28-unit mapping table; units not in the table produce missing values — no guessing or capping. Binary survey fields (`land_irrigated`, `foodshortagetime`) were encoded as numeric flags. HFIAS and FIES strings were retained as-is; "no_answer" is a valid response category, not a missing value.

**No imputation was applied.** Missing values remain missing throughout. Missingness in RHoMIS reflects survey design — which questions appeared in which waves, and which households answered them. Filling with means or zeros would misrepresent the data.

**Feature engineering** (analysis phase — notebooks 04 and 05):

| Derived Field | Formula / Logic |
|---|---|
| `low_education_flag` | $= 1$ if `education_level` $\in$ {no formal, primary}; else 0 |
| `crop_diversity_count` | $= \lvert\{\texttt{crop_name_}i : i = 1 \ldots 5,\ non-null\}\rvert$ per household |
| `land_productivity_kg_per_ha` | $= \sum_{i=1}^{3} \texttt{crop_harvest_kg_}i\texttt{land_cultivated_ha}$ |
| `country_income_rank_pct` | $= percentile_rank(\texttt{total_income_ppp})$ within each country |
| `vulnerability_profile_k4` | $k$-means ($k = 4$) on food security, farm structure, and income indicators |
| `productivity_quartile` | Q1–Q4 equal-frequency bins over non-missing `land_productivity_kg_per_ha` |

## 6. KPI & Metric Framework

*Reference: notebooks/05_final_load_prep.ipynb, Section 4*

Seven KPIs form the backbone of the dashboard. Each is interpretable without statistical training and directly connected to at least one recommendation.

| KPI | Value | Formula | Business Purpose |
|---|---|---|---|
| Food Shortage Rate | 67.7% | $\frac{HHwithshortage}{HHwithvalidresponse}$ | Headline vulnerability metric. Baseline for all targeting. |
| Food-Shortage HH Count | 32,097 | Count(`food_shortage_flag` = 1) | Translates the rate into a countable population. |
| Median Months Shortage | 3.0 mo | median(`foodshortagetime`) | Severity indicator — seasonal lean-period risk. |
| Irrigation Access Rate | 20.2% | $\frac{irrigated_flag=1}{totalHH}$ | Linked to 16.3pp food shortage advantage on D3. |
| Median Land Productivity | 600 kg/ha | median(`land_productivity` | Farm performance baseline. Quartiles segment risk. |
| Median Income per MAE | $112 PPP | median(`income_ppp_per_mae`) | Welfare baseline, PPP-normalised cross-country. |
| Shortage — Lowest Quintile | 75.8% | avg(`food_shortage_flag`) where quintile = 1 | Bottom quintile faces highest structural risk. |

## 7. Exploratory Data Analysis

*Reference: notebooks/03_eda.ipynb committed to GitHub*

### 7.1. Food Shortage Distribution

Of 47,399 households with valid responses, 67.7% reported food shortage in the past year. Country-level variation is extreme: the Democratic Republic of Congo leads at 94.5% (among countries with ≥500 households), followed by Rwanda (94.3%), Comoros (90.7%), and Malawi (89.5%). At the other end: India (10.5%), Palestine (14.9%), Nepal (18.1%), and Costa Rica (18.6%). This 84-point spread confirms that country-level context — conflict, infrastructure, climate — is a major moderating factor.

### 7.2. Structural Drivers

*Irrigation* creates the clearest outcome gap: irrigated households face a 55.2% shortage rate vs. 71.5% for rainfed — a 16.3pp difference — yet only 20.2% of households have access. *Land productivity* shows the widest spread: Q1 households face 83% shortage vs. 54% for Q4, a 29.1pp gap. *Income position* contributes a 19.8pp gap between the bottom and top income quintiles within each country. *Crop diversity* shows a positive correlation with food security in EDA but weakens significantly after controlling for income — suggesting the effect is largely mediated by household wealth.

### 7.3. Food Security Instrument Validation

Moving from 0 to 8 affirmative FIES responses corresponds to a 72.1pp increase in observed food shortage rate in the 24,093 households where both measures exist — confirming the self-report question and the FIES instrument capture the same underlying condition.

### 7.4. Gender

Female respondents show a marginally higher food shortage rate (Cramér's V = 0.078). Median female agricultural value control is near parity at 0.5, though this aggregated figure masks significant within-country variation.

## 8. Statistical Analysis

*Reference: notebooks/04_statistical_analysis.ipynb committed to GitHub*

### 8.1. Categorical Driver Tests

Chi-square tests with Cramér's V confirm all four categorical drivers are independently associated with food shortage at $p < 10^{-30}$:

| Driver | Chi-square | Cramér's $V$ | Interpretation |
|---|---|---|---|
| Low education | 874.1 | 0.145 | Strongest categorical predictor |
| Irrigation access | 869.5 | 0.138 | Near-equal to education |
| Respondent sex | 281.5 | 0.078 | Significant, weaker effect |
| Off-farm income | 152.2 | 0.057 | Consistent, weakest of four |

### 8.2. Numeric Associations

Mann-Whitney U tests were used (parametric tests inappropriate given extreme skew). Spearman's $\rho$ for land productivity vs. food shortage flag: $-0.240$ ($p \approx 0$). FIES validation: Spearman $\rho = 0.587$ (n = 24,093, $p \approx 0$). HFIAS validation: Mann-Whitney effect = 0.334 (n = 5,911, $p < 10^{-170}$).

### 8.3. Multivariable Regression

Country- and year-controlled logistic regression on 34,927 complete households, with log transformations applied to skewed continuous predictors:

| Predictor | Odds Ratio | 95% CI | Interpretation |
|---|---|---|---|
| Within-country income rank | 0.456 | [0.408, 0.510] | Strongest protective effect |
| Log cultivated area | 0.671 | [0.637, 0.706] | Larger land reduces odds |
| Log land productivity | 0.828 | [0.810, 0.845] | Higher productivity reduces odds |
| Low education flag | 1.268 | [1.184, 1.358] | 27% higher odds of shortage |
| Household size | 1.011 | [1.006, 1.017] | Small additive risk per member |

### 8.4. Vulnerability Segmentation

k-means clustering (k=4 selected via silhouette score) on food security, farm structure, and income indicators identifies four distinct household profiles:

| Profile | HH | Shortage | Irrigated | Key Characteristics |
|---|---|---|---|---|
| Low-income rainfed | 14,330 | 78.7% | 0.3% | Smallest farms, lowest income (rank 0.24) |
| Large-HH extensive | 6,860 | 72.3% | 4.4% | 13-member average, extensive land, low productivity |
| Diversified better-off | 12,855 | 62.7% | 0.0% | Higher income rank (0.70), more crop diversity |
| Irrigated high-prod. | 7,753 | 55.8% | 100% | Lowest risk; irrigation is the distinguishing factor |

*13,075 households unclassified due to insufficient data across clustering variables.*

## 9. Tableau Dashboard Design

*Reference: Published at public.tableau.com — workbook: RHoMIS DASHBOARD. Screenshots below; source file in `tableau/` on GitHub.*

### 9.1. Objective

The system supports a single decision: where should a development organisation focus its next intervention, and what structural factor should it target? Four interconnected dashboards move a user from geographic overview to farm-profile detail to structural driver analysis to income position. Cross-dashboard navigation buttons on every view let users move through the system without returning to the Tableau menu.

### 9.2. Structure

| Dashboard | Purpose and Key Charts |
|---|---|
| D1 — Risk Overview | Choropleth world map of food shortage rate by country; horizontal bar chart of shortage rate ranked by country; lollipop chart of median months of food shortage; five KPI summary cards |
| D2 — Vulnerability Profiles | Grouped bar chart of food shortage rate by vulnerability profile and country; pie chart of profile distribution; demographic indicator heatmap by profile (irrigation rate, low-education rate, female respondent rate, off-farm income rate) |
| D3 — Structural Drivers | Side-by-side bar chart of food shortage rate by productivity quartile and irrigation type; bar chart of food shortage rate by crop diversity count; scatter plot of land productivity vs. food shortage rate by country |
| D4 — Income Position | Scatter plot of income rank vs. food shortage rate per country, coloured by vulnerability profile; grouped bar chart of median income by profile with and without off-farm income; diverging bar chart of income quintile distribution by country |

All dashboards include Country and Vulnerability Profile filters. D1 additionally filters by Irrigated Flag, Respondent Sex, and Year. Reference lines on scatter charts use fixed constants — global food shortage average 67.7% and country income median 0.5 — not dynamic averages that shift under filtering.

### 9.3. Screenshots

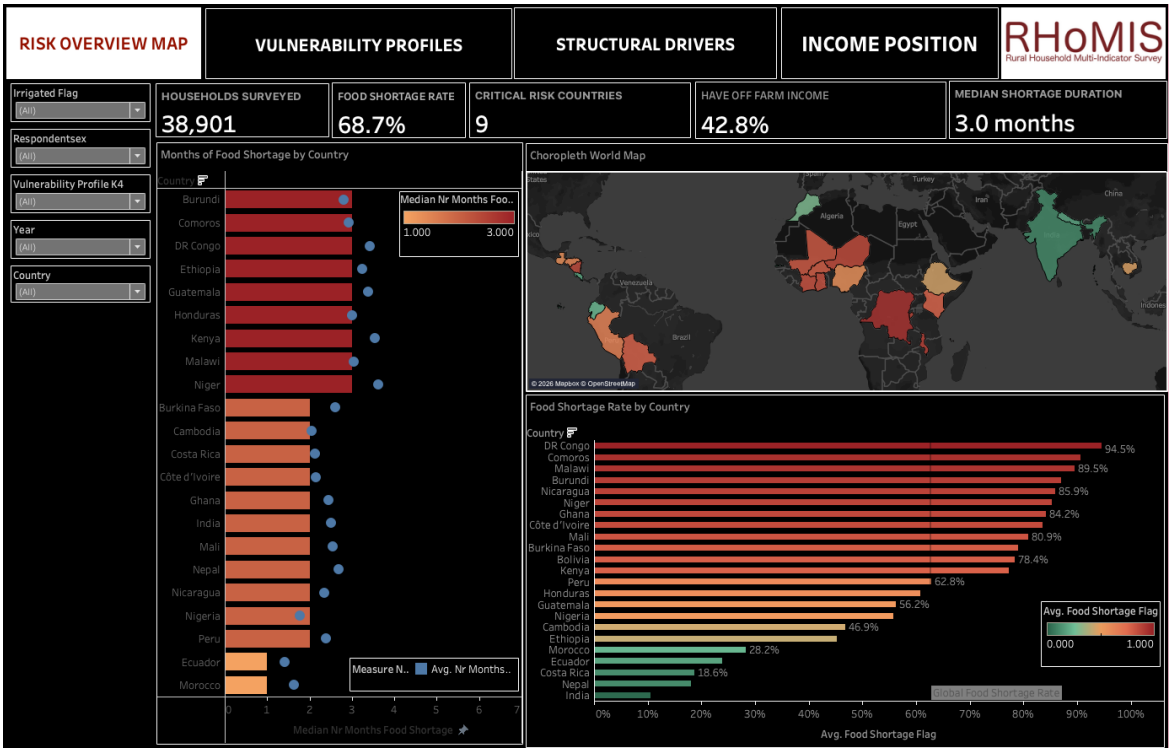


**Figure 1.** D1 — Risk Overview: choropleth map, food shortage rate by country, and median months of shortage.

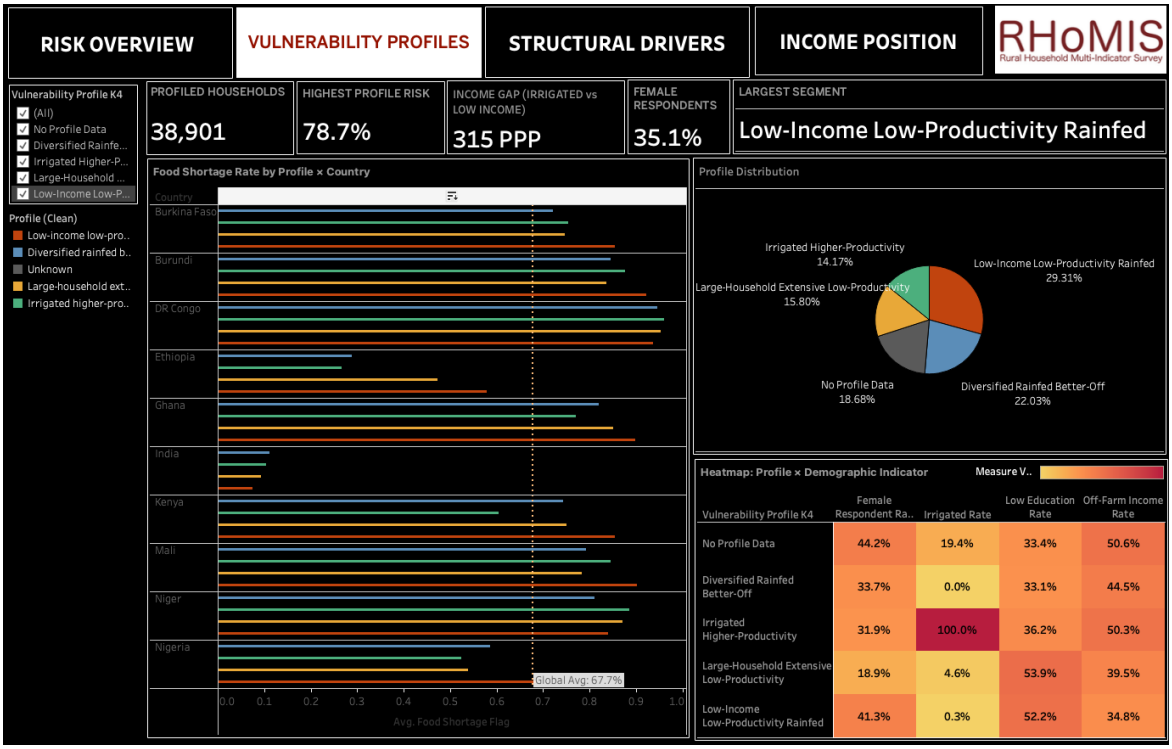


**Figure 2.** D2 — Vulnerability Profiles: food shortage rate by profile and country, profile distribution, and demographic indicator heatmap.

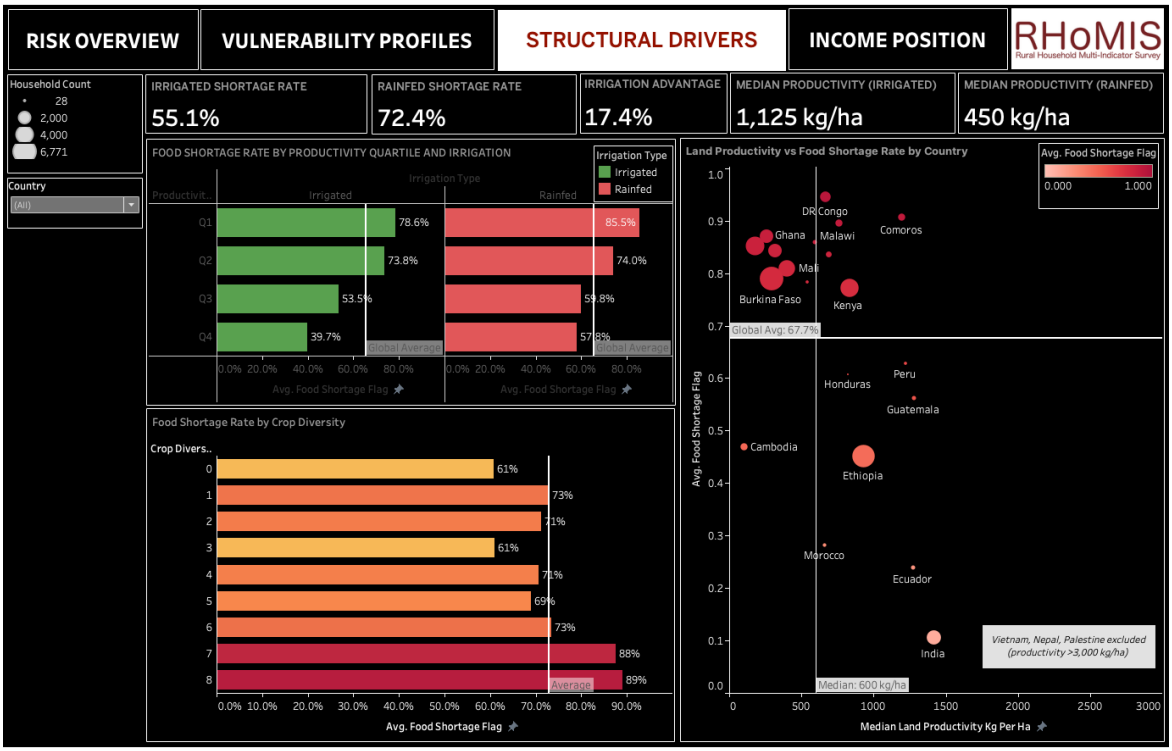


**Figure 3.** D3 — Structural Drivers: productivity quartile by irrigation type, crop diversity bar, and land productivity scatter by country.

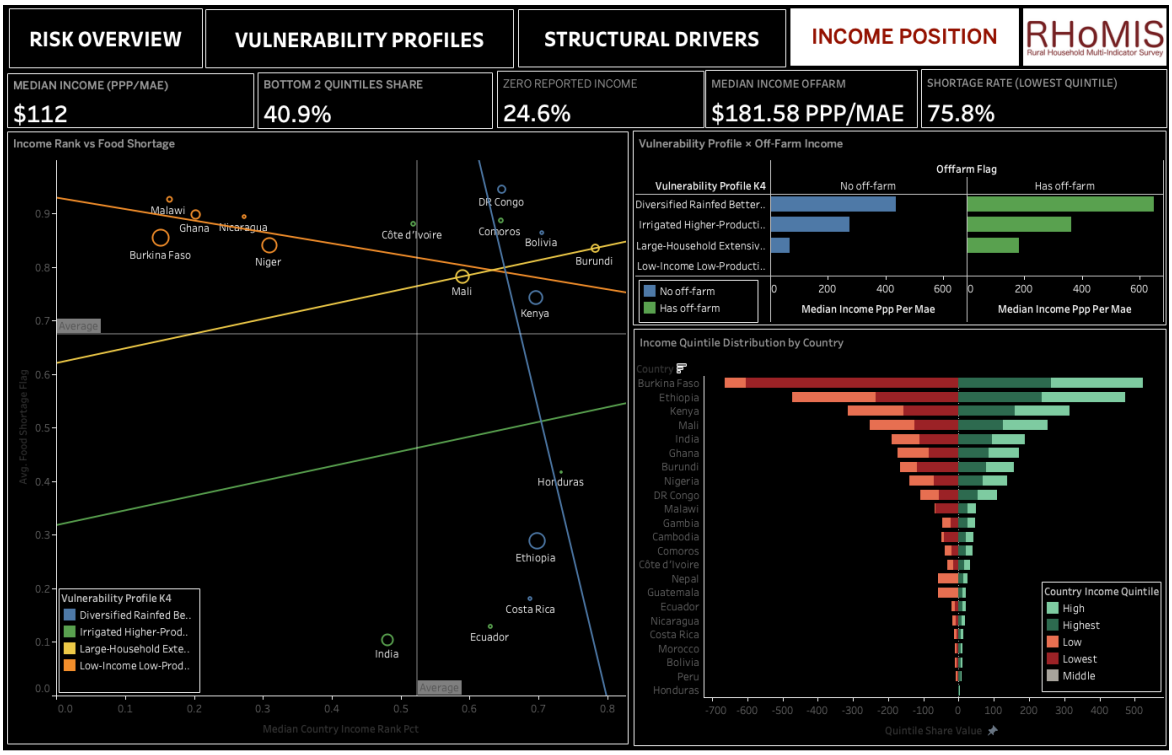


**Figure 4.** D4 — Income Position: income rank vs. food shortage scatter, off-farm income comparison by profile, and income quintile distribution by country.

**Tableau Public URL:**

`https://public.tableau.com/app/profile/rajdeep.sanyal/viz/RHoMISDASHBOARD/INCOMEPOSITION`

## 10. Insights Summary

1. **Food insecurity is the structural norm, not the exception.** A 67.7% global rate means that programmes designed on the assumption that most farms are food-secure will miss their target population entirely.
2. **Irrigation offers the largest addressable gap.** A 16.3pp food shortage difference against 20.2% coverage means infrastructure investment has both a large effect size and a large addressable population.
3. **Land productivity is the deepest structural divide.** The 29.1pp gap between Q1 and Q4 productivity quartiles is an independent risk factor — not a proxy for country poverty or income.
4. **Income position is the single strongest predictor.** In the controlled regression, income rank has the strongest protective odds ratio (0.456). Addressing income alone — without any farm structural change — can move the food security outcome.
5. **Low education amplifies every other risk.** With the highest Cramér's V (0.145) and an odds ratio of 1.268, low education functions as a vulnerability multiplier across all other predictors.
6. **DRC, Rwanda, Senegal, and Comoros need immediate priority.** All four exceed 90% food shortage rates. Country-level factors — conflict, infrastructure, climate — carry substantial explanatory weight beyond farm structure.
7. **Low-risk countries prove improvement is possible.** India (10.5%), Palestine (14.9%), Nepal (18.1%), and Costa Rica (18.6%) demonstrate that smallholder food insecurity is not an inevitable condition.
8. **The most vulnerable cluster has almost no irrigation.** Low-income rainfed households (14,330, 78.7% shortage) have 0.3% irrigation coverage — the most concentrated intervention target in the dataset.
9. **Extensive land does not protect against food insecurity.** The large-household extensive cluster holds more land than any other profile yet records a 72.3% shortage rate. Land area without productivity is not protective.
10. **Crop diversity requires income support to be effective.** Its correlation with food security weakens after income controls — diversification programmes work best when bundled with income access, not delivered in isolation.
11. **A three-month median duration points to seasonal targeting.** Lean-season food storage bridges and income smoothing programmes are likely more cost-effective than general food assistance.
12. **Vulnerability is structurally predictable.** The five-variable regression model correctly separates food-secure and food-insecure households 81% of the time, confirming that targeting based on observable farm characteristics is statistically viable.

## 11. Recommendations

**Rec 1 — Irrigation expansion in the low-income rainfed cluster**

*Insight:* 16.3pp food shortage gap between irrigated and rainfed; only 0.3% irrigation in the highest-risk cluster (14,330 HH, 78.7% shortage).

*Action:* Prioritise small-scale and drip irrigation in DRC, Rwanda, Comoros, and Malawi — starting with the low-income rainfed cluster.

*Impact:* Estimated 2,300 households per 1,000 reached exit acute food insecurity if coverage moves from 0.3% to 20%.

*Feasibility:* Requires water-source audits and infrastructure capital. Timeline: 12–36 months.

**Rec 2 — Productivity support for Q1 and Q2 farms**

*Insight:* 29.1pp gap between productivity quartiles; confirmed as independent driver in regression.

*Action:* Deploy extension services (improved seed varieties, soil management) to the bottom two productivity quartiles — approximately 24,000 households.

*Impact:* Estimated 10–15pp shortage rate reduction for Q1 households reaching Q2 productivity levels.

*Feasibility:* Low-cost, fast-deployable via existing extension networks. Timeline: within 6 months.

**Rec 3 — Income diversification and education for the bottom quintile**

*Insight:* Income rank is the strongest predictor (OR 0.456); low education amplifies all other risks (OR 1.268); both compound in the lowest quintile.

*Action:* Integrate off-farm income pathways (rural microenterprise, remittance access) and adult literacy for bottom-quintile, low-education households.

*Impact:* Estimated 4–8pp shortage reduction per income quintile step.

*Feasibility:* Income support via microfinance within 3–6 months; education requires government coordination.

## 12. Impact Estimation

| Intervention | Target Population | Estimated Effect | HH Exiting Insecurity (per 1,000 reached) |
|---|---|---|---|
| Irrigation expansion | 14,330 cluster | rainfed | −16.3pp | ≈2,300 |
| Productivity support | 24,000 farms | Q1+Q2 | −10–15pp | ≈1,200–1,800 |
| Income + education | 10,200 quintile | bottom | −4–8pp | ≈400–800 |

Estimates are derived from quartile and quintile comparisons in the EDA and should be treated as directional, not causal. The urgency case: 32,097 households are currently in food shortage, with structural drivers that are identifiable and addressable through existing programme types.

## 13. Limitations

- **Cross-sectional design.** Each household appears only once. All findings are associations, not causal effects. Intervention impact estimates are group comparisons, not experimental results.
- **Self-reported outcomes.** Food shortage status is subject to recall and social desirability bias. FIES and HFIAS validation supports reliability, but objective measurement was not available.
- **Income data artefacts.** Extreme income outliers (up to 350 billion PPP) arise from data entry errors in multi-crop income fields. All income analysis uses within-country rank, which is robust to absolute outliers.
- **Uneven year distribution.** 2018 = 22% of sample; 2015 = 2%. Year-over-year trend analysis excluded. Year controlled as a fixed effect in regression only.
- **Region names unstandardised.** Sub-national comparison below country level is not possible without consistent administrative naming.

## 14. Future Scope

- **Longitudinal panel linkage.** Tracing the same households across survey years would enable within-household change estimates and move toward causal inference.
- **Predictive targeting tool.** The five-variable model is strong enough to serve as a lightweight household vulnerability screening tool applied to new survey waves, reducing targeting cost without a full analytical refresh.
- **Climate data integration.** Linking GPS coordinates to satellite-derived rainfall, NDVI, and temperature anomaly data would isolate structural farm characteristics from year-specific climate shocks.
- **Sub-national disaggregation.** Standardised region names would unlock district- or province-level analysis within countries.

## 15. Conclusion

Analysis of 54,873 smallholder households across 35 countries confirmed that food insecurity is pervasive (67.7%), geographically concentrated in Central and West Africa, and structurally driven by a small number of addressable factors. Income position, cultivated land area, land productivity, and low education emerge as the four most important predictors in a country-controlled regression. The clearest single action is irrigation expansion: reaching the 79.8% of households currently without it — beginning with the low-income rainfed cluster at 78.7% shortage and 0.3% irrigation coverage — is associated with a 17-percentage-point improvement in food security probability. Development organisations that concentrate irrigation, productivity support, and income diversification on the highest-risk cluster will reach the largest food-insecure population with the highest-return combination of evidence-based interventions.

## A. Data Dictionary

Full column-by-column reference for all 46 fields in `rhomis_tableau_ready.csv` is maintained in `docs/data_dictionary.md` in the GitHub repository, covering type, missing rate, value range, and analytical caveats for every field across seven thematic groups.

## B. Contribution Matrix

| Name | Dataset | ETL | EDA | Stats | Tableau | Report | PPT/Viva |
|---|---|---|---|---|---|---|---|
| Mahir Abdullah | ✓ | | ✓ | ✓ | ✓ | ✓ | |
| Manas Vivek Saxena | ✓ | ✓ | | | | | ✓ |
| Manya Verma | ✓ | ✓ | | | | | |
| Rajat Srivastav | ✓ | ✓ | | | | | |
| Rajdeep Sanyal | ✓ | | | | ✓ | | |
| Sankalp | ✓ | | | ✓ | ✓ | | |

**Team Lead:** Mahir Abdullah **Date:** 28 April 2026

*Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.*